

Day 3 :- Combinational Assignment

1) Implement the following functions using (i) 2:1 mux and NOT gates (ii) 4:1 mux

a) $F(A, B, C) = \bar{A}C + B\bar{C}$

choose C as select line

For 2:1 mux

$$F = \bar{S}I_0 + SI_1$$

When $C = 0$, $F = B$

When $C = 1$, $F = \bar{A}$

Hence $S = C$, $I_0 = B$, $I_1 = \bar{A}$

$$F = \text{MUX}(C, B, \bar{A})$$

b) using 4:1 mux

Take A, B as select lines

A	B	F
0	0	C
0	1	1
1	0	0
1	1	$\bar{C}$

$$S_1 = A, S_0 = B$$

$$I_0 = C, I_1 = 1, I_2 = 0$$

$$I_3 = \bar{C}$$

$$F = \text{MUX}(A, B, C, 1, 0, \bar{C})$$

2) Implement the functionality of a full adder using 2:1 mux only

$$S = A \oplus B \oplus \text{cin}$$

$$\text{Cout} = AB + B\text{cin} + A\text{cin}$$

Using 2:1 mux with select A

$$A \oplus B = \bar{A}B + A\bar{B}$$

Select - A

$$I_0 = B$$

$$I_1 = \bar{B}$$

$$X = A \oplus B$$

$$S = (A \oplus B) \oplus \text{cin}$$

$$X = A \oplus B$$

$$S = X \oplus \text{cin}$$

Using 2:1 mux

X $\rightarrow$ select line

$$S = \bar{X}(\text{cin}) + X(\bar{\text{cin}})$$

$$S = X$$

$$I_0 = \text{cin}, I_1 = \bar{\text{cin}}$$

$$S = (A \oplus B) \oplus \text{cin}$$

Carry :-

$$\text{Cout} = AB + A\text{cin} + B\text{cin}$$

$$F = \bar{S}I_0 + SI_1$$

$$X = A \oplus B$$

$$X = 0$$

$$A = B$$

$$A \quad B$$

$$0 \quad 0$$

$$0 \quad 1$$

$$\text{When } A = B = 0$$

$$\text{cout} = 0$$

$$\text{When } A = B = 1, \text{ cout} = 1$$

$$\text{Whenever } X = 0$$

$$\text{cout} = A$$

$$X = 1$$

$$A \neq B$$

$$A \quad B$$

$$0 \quad 1$$

$$1 \quad 0$$

$$\text{When } A = 0, B = 1$$

$$\text{cout} = 0 + (0)\text{cin} + 1 \cdot \text{cin} = \text{cin}$$

$$\text{When } A = 1, B = 0$$

$$\text{cout} = 0 + 1 \cdot \text{cin} + 0 = \text{cin}$$

$$\underline{\underline{\text{cout} = \text{cin}}}$$

$$Y = \bar{3}I_0 + 8I_1$$

$$Y = X = A \oplus B$$

$$X = 0 \Rightarrow A$$

$$X = 1 \Rightarrow \text{cin}$$

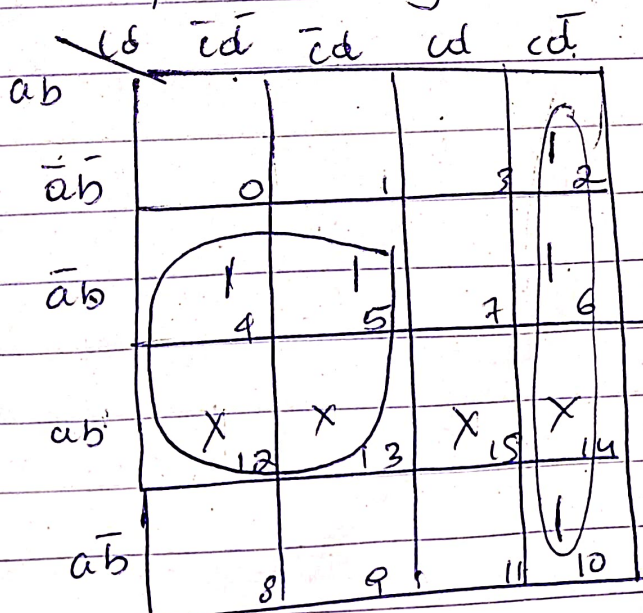
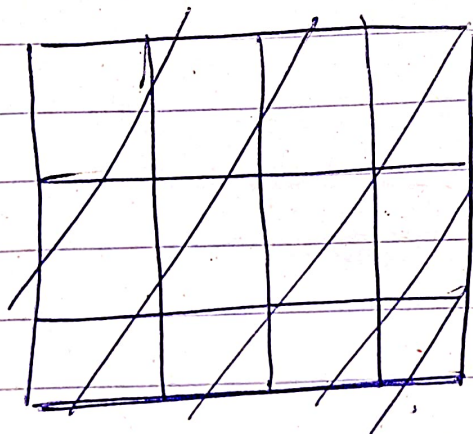
$$I_0 = A$$

$$\text{cout} = \bar{X}A + X\text{cin}$$

$$I_1 = \text{cin}$$

$$\text{cout} = (\bar{A} \oplus B)A + (A \oplus B)\text{cin}$$

3) Simplify $f(a, b, c, d) = \sum m(2, 4, 5, 6, 10) + \sum d(12, 13, 14, 15)$ and implement the sop expression using 4:1 mux



$$\underline{\underline{b\bar{c} + c\bar{d}}}$$

⑥ Implement using 4:1 mux

Choose B and C as select lines

$$S_1 = B \quad S_0 = C$$

$$BC = 00 \quad \text{then } F = 0 \Rightarrow I_0 = 0$$

$$BC = 01 \quad \text{then } F = D' \Rightarrow I_1 = D'$$

$$BC = 10 \quad \text{then } F = 1 \Rightarrow I_2 = 1$$

$$BC = 11 \quad \text{then } F = D' \Rightarrow I_3 = D'$$

$$\underline{F = BC' + CD'}$$